

Deliverable D - Technical Writeup

Team: Hossain Pazooki

What ships vs. what we propose (read first). Our submission is a **sklearn-only baseline that passes the official validator**: a weekly **competing-risks discrete-time hazard ensemble** (HistGradientBoostingClassifier, 5 seeds) -> a calibrated lifetime PD -> an **upper-bound NPV approval rule** (A), a cohort cumulative-incidence trajectory (B), and $\text{do}(\cdot)$ counterfactuals with structural propagation (C); intervals are **ensemble percentile bands**, recalibrated **out-of-time** with isotonic regression. The doubly-robust off-policy, IPW reweighting, split-conformal, and DeepHit machinery discussed below is **proposed and was not built** - and we explain why one of them (IPW) is not merely unbuilt but *unidentifiable* here. The strongest contribution is not model weight; it is a disciplined account of *what is identifiable, what is partially identifiable, and what is pure extrapolation* in this data, and a baseline that acts accordingly.

1. Problem framing & assumptions violated

The lender maximizes realized portfolio value. With no binding capital constraint the objective is separable across applicants, so the optimal policy is **pointwise on the sign of expected NPV**, $d^*(x) = 1\{E[\text{NPV} \mid x, \text{do}(\text{fund})] > 0\}$. The payoff is **sharply asymmetric**: a repaid loan returns single-digit percent of principal while an early default with low recovery destroys most of it. The task is therefore to estimate a **calibrated expected NPV**, not to maximize classification accuracy - decisions flip in a narrow band, so a few points of miscalibration destroy value.

Four data realities - all recomputed from the CSVs - break standard assumptions:

(a) Selective labels with a deterministic funding rule. Repayment outcomes exist only for funded, matured loans (**60.6%** of train, 51,722/85,340). The legacy funding decision is a **perfect deterministic threshold** on an observed score: `prior_decision == 1` iff `prior_underwriter_score >= ~0.273`, with **zero mismatches across all 85,340 rows** and a clean gap (max declined 0.27297 < min approved 0.27301). So the funding propensity $e(x) = P(\text{fund} \mid x)$ is **degenerate (0/1) and positivity fails globally** - there is no overlap region. This is the spine of the submission and is developed in Section 3.

(b) Out-of-time deployment. Training spans 2024-01 to 2025-06; the decision population (validation + test, **13,306** applicants) is a single later 13-week window (2025-06-30 to 2025-09-28). We evaluate **out-of-time** on the validation split, mirroring deployment - not a random split. Validation labels also exist only for its **funded** subset (2,551/4,489), so even calibration data is policy-selected.

(c) Path-dependent timing with a point mass. Default triggers on missed-draw rules or a nonzero balance at day 90. Verified timing: all paid-in-full loans repay at **exactly day 60**; defaults occur over days 3-60 **and then jump to a point mass at exactly day 90** (**22.5%** of defaults), with **zero defaults in (60, 90)**. A single life-of-loan flag discards timing that both the NPV and Deliverable B need.

(d) Informative missingness (MNAR by design). The bank-feed block is null iff `has_linked_bank_feed == False` (exact across all rows); `days_since_*` nulls mean "no prior event." We preserve missingness as signal (indicator flags + native NaN handling), never blind-impute.

2. Methodology

One fit, three deliverables. Rather than regress NPV directly (high variance, opaque), we model only its stochastic inputs and compose them through the known cash-flow algebra. A **single weekly competing-risks discrete-time hazard** over the two absorbing events - **default vs payoff** - is fit on funded+matured loans in person-period layout (HistGradientBoostingClassifier, native NaN + sample-weight, 5 seeds bagged). Survival $S(t) = \prod_{s \leq t} (1 - h_d(s))$ and cumulative incidence $\text{CIF}_d(t) = \sum_{s \leq t} h_d(s) S(s-1)$ give lifetime PD = $\text{CIF}_d(H)$ for A and the full trajectory for B off **one coherent fit**. Carrying the payoff hazard prevents overstating PD (a loan that pays off is removed from default risk). The economics: amount R in \$5-50K, 60-day term, 35% APR, 3% fee, 90-day default window -> $E[\text{NPV} \mid x] = (1 - \pi)(F + \text{interest}) + \pi \cdot E[\text{recovery} - \text{loss}]$, with $F = 0.03R$.

Interest convention, disclosed not assumed. Reading "35% APR" as interest on full principal for the 60-day term gives $R \cdot 0.35 \cdot 60 / 365 \approx 0.0575R$, so a repaid loan returns $F + \text{interest} \approx 0.0875R$; the amortizing (declining-balance) reading roughly halves the interest. We **state the full-principal convention explicitly** because the headline break-even PD swings materially between them.

Break-even is a band, not a point. $NPV(p_i)$ is linear and strictly decreasing in p_i ; setting it to zero, $p_i^* = g / (g + 1 - \text{rec})$. With $g = 0.0875$: zero recovery gives $p_i^* \approx 8.05\%$, empirical recovery ($\approx 0.091R$) gives $\approx 8.8\%$ - a band $\approx 8.0\text{-}8.9\%$ (the amortizing reading gives $\approx 5.5\text{-}6.1\%$). The funded-book default rate **17.45%** sits well above this band, confirming the legacy book funds many negative-NPV loans - the headroom our policy targets.

Decision rule - fund on the conservative bound. Because NPV decreases in PD, the pessimistic case is the **highest** credible default rate, so we **approve only if NPV stays positive at the UPPER 90% PD bound**. Uncertain loans push their upper-bound PD past break-even and **default to denial** - ignorance is denied, not approved (the inverse rule would be backwards). On the 13,306 applicants this yields **7.2% approval** (957 loans), approved-set mean PD **0.041** with **every approval's upper-bound PD < 0.084 < break-even**, vs declined-set mean PD **0.28**.

Feature hygiene. The outcome model excludes `prior_underwriter_score` (Section 3), `prior_approved_amount` (funded-only leakage), and the constant `prior_decision`; features are kept **numeric with NaN preserved** (no category dtype) to avoid cross-split alignment bugs. Recovery is an **empirical mean $\approx 0.091R$** with disclosed dispersion (median 0.072, $\approx 23\%$ recover exactly 0), not yet a censoring-aware regressor. *Proposed, not built*: a broader LightGBM/XGBoost/logistic model family (the stack is deliberately sklearn-only for runtime portability) and a Markov missed-draw simulation of the exact default automaton.

3. Causal reasoning & counterfactual methodology

Deliverable C asks for an **interventional** quantity, $P(\text{default} \mid \text{do}(f=v), X_{\{-f\}} = x_{\{-f\}})$, which differs from the **observational** $P(\text{default} \mid f=v, \dots)$ whenever f is confounded. Re-scoring a fitted predictor with one feature swapped - and, equivalently, reading a SHAP attribution - answers the *observational* question and is **not** a do-operator estimate. We state this plainly rather than dressing a model perturbation as a causal effect.

The identification problem is the core finding. Repayment is observed only on the historically funded set S , selected by the legacy policy. We verified that policy is a **deterministic threshold** on `prior_underwriter_score` (0 mismatches/85,340; no overlap), so the funding propensity is **degenerate and positivity fails globally**. The consequences are concrete and we follow each to its honest end:

- ****Covariate-shift / IPW reweighting toward the applicant marginal is not**

identified - there are no funded look-alikes at a declined applicant's score to reweight. We therefore removed the IPW / reject-inference path entirely **and fit the hazard** unweighted on funded labels only^{**}; `propensity.py` survives purely as a diagnostic that *proves* the rule and reports positivity. Demoting IPW from "a thing we do" to "the failure mode that forces abstention" is deliberate honesty.

- **`prior_underwriter_score` is excluded from the outcome model.** Funded rows only ever carry `score >= 0.273`, yet **43.5%** of the decision population sits *below* that minimum - the score is out of training support for nearly half the applicants we must decide on. Using it would extrapolate a legacy-policy artifact across the funding boundary and bake selection into the default model.
- **Outside funded support we do not extrapolate silently.** Any cross-threshold prediction for a declined-type applicant is an explicit modeling extrapolation (the default function assumed smooth across the boundary); we widen intervals, use **Manski-style partial-identification** language, and abstain where appropriate.

What the baseline does honestly for C. Where a dependency is *deterministic* we **propagate structurally**: an intervention on `has_linked_bank_feed` propagates to the entire bank-feed block (verified strict gating), not a lone boolean; interventions on `requested_amount` / `revenue` propagate to the engineered `requested_amount_to_observed_revenue` ratio. These are mechanical, defensible edits. **Everything else is reported as the model's observational response under feature substitution, with the observational-vs-interventional gap surfaced, not hidden.** The query set is 900 queries / 300 applicants / 30 features; we distinguish **hard-structural identity** features (sector, region, vintage, employee bucket - not coherently intervenable) from **merely-not-actionable history** (account age, prior loan counts). For confounded drivers like bank-feed linkage we make the confounding case via **covariate association** (linkers vs non-linkers differ in credit utilization ≈ 0.40 vs 0.47 , requested amount, and underwriter score), **not** via raw default rates - which are nearly identical (0.207 vs 0.206) and would *understate* confounding. C intervals are correspondingly **wider than A** (mean width 0.091 vs 0.070), reflecting that interventions are less identified.

Regulatory defense of stated drivers. For each driver we would argue **sign, magnitude, and mechanism**, and explicitly flag proxies as associational unless backed by the structural-propagation logic above. The honest regulator-facing claim is: "this feature moves the model's *predicted* risk by X; we can defend it as a *causal* lever only where the dependency is mechanical (bank-feed gating, ratio recomputation); elsewhere it is an association, and for applicants below the legacy funding threshold it is an extrapolation we bound rather than assert." *Proposed, not built*: a backdoor-adjusted SCM / Double-ML / causal-forest estimator - under global positivity failure several interventions are only partially identified, so we prioritized stating the gap over shipping a point effect that looks solved.

4. Calibration & uncertainty quantification

Calibration is the scoreboard, and is applied out-of-time. Because decisions flip in a narrow band near $\pi^* \approx 8.8\%$, ranking is secondary to calibration. We fit an **isotonic** recalibrator on the **validation funded subset** ($n=2,551$ - its realized lifetime default, scored by the train-fit ensemble, a genuinely out-of-time check) and **apply it to the submitted A/B/C PDs**. The map is monotone, so it preserves lower $\leq$ point $\leq$ upper and B's per-cohort age-monotonicity, and B is calibrated with the **same** map as A so the cross-check `cdr@age13` $\approx$ approved-set PD holds (0.047 vs 0.041). We gate the application on a **K-fold cross-fitted** ECE (held-out, never isotonic's degenerate in-sample ≈ 0): out-of-time ECE improves **$0.0228 \rightarrow 0.0199$** and mean predicted PD rises **$0.187 \rightarrow 0.206$** to match realized **0.206** . The decision effect is large precisely because the model under-predicts *near break-even*, which tightens approvals from **$\sim 21\%$ (uncalibrated) to $\sim 7\%$** - the correct conservative consequence of honest calibration. **Caveat, stated**: the calibration slice is *funded*, so applying the map across the whole population rests on a smoothness/extrapolation assumption - concentrated where it is most trustworthy (the in-support, near-break-even region).

Intervals. 90% bands are the **5th/95th percentiles of the 5-seed hazard bag** for A, B, and C. We **fund on the conservative bound** (upper PD = lower NPV), so ignorance defaults to denial and bands widen near the support boundary. We chose ensemble dispersion over a single point because the binding uncertainty here is *model/extrapolation* uncertainty across an unobserved region, which seed dispersion exposes; the trade-off is that these bands are **honest-but-not-coverage-guaranteed**. *Proposed, not built*: **split-conformal** intervals for finite-sample coverage (`calibration.conformal_intervals` exists but is unused) - the principled upgrade, though under the positivity failure even conformal coverage holds only in-support.

5. Limitations & what we'd do differently

- **Identification, not estimation, is the binding constraint.** The deterministic funding rule means no reweighting recovers declined-applicant outcomes; we bound and abstain rather than ship a confident-looking but unidentified IPW/DR number. This is the submission's deliberate stance, not an omission.
- **The calibration and counterfactual maps lean on a smoothness assumption** across

the funding boundary (positivity fails). With another day we would quantify its fragility - sensitivity of approvals to the assumed cross-boundary slope, and Manski bounds on the approved-set value.

- **Recovery is observed only on a selected sliver** of defaulters and modeled as a shrunken mean; a censoring-aware recovery regressor is the natural next step.
- **Intervals are ensemble percentile bands, not conformal** - coverage is plausible, not guaranteed; split-conformal (in-support) is the first upgrade.
- **Off-policy value is only partially identified** here; a doubly-robust $V(d)$ with bootstrap CIs and a Rosenbaum sensitivity check is the evaluation we would *report as bounds*, not a single number, and did not build.
- **Stack is deliberately sklearn-only** (HistGradientBoosting + IsotonicRegression, pandas/numpy); the hazard spine is model-agnostic, so swapping in a richer learner family (LightGBM/XGBoost/logistic stack) is a clean, proposed upgrade path.

What we are most confident in: the data diagnosis (every empirical figure here is recomputed from the raw CSVs and survived an adversarial verification pass), the decision-rule direction (conservative upper-bound), and that A/B/C pass the official validator (RESULT: PASS, 0 errors). What we are least confident in: any prediction for applicants below the legacy funding threshold - which is exactly why we bound it.

References. Lakkaraju et al. (2017), *The Selective Labels Problem*, KDD. Kozodoi et al. (2025), *Fighting Sampling Bias*, EJOR 324(2):616-628. Lessmann et al. (2015), EJOR 247(1):124-136. Verbraken, Verbeke & Baesens (2013), *Profit-maximizing metric*, IEEE TKDE 25(5):961-973. Bellotti & Crook (2013), *Forecasting and stress testing credit card default*, IJF 29(4):563-574. Angelopoulos & Bates (2023), *Conformal Prediction: A Gentle Introduction*, FnT ML 16(4):494-591. Athey & Wager (2021), *Policy Learning With Observational Data*, Econometrica 89(1):133-161. Lundberg & Lee (2017), *A Unified Approach to Interpreting Model Predictions*, NeurIPS (with the SHAP non-causal caveat). Pearl (2009), *Causality* (2nd ed.), Cambridge UP.